Machine learning modeling for high critical temperature superconductor discovery

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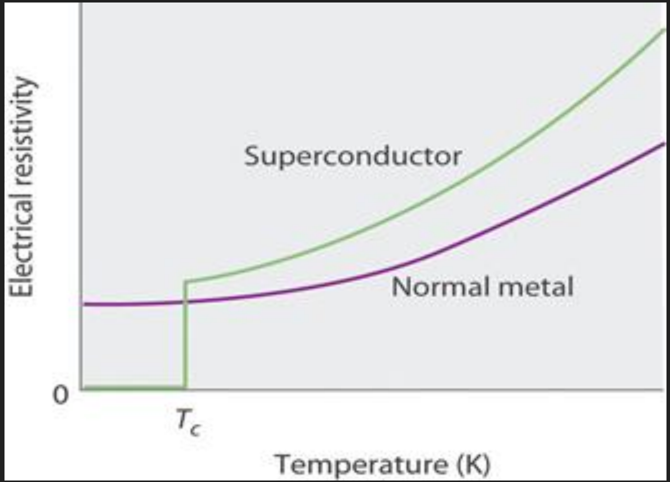
https://github.com/drooryck/superconductors-ml

Motivation: Superconductors!



What is a superconductor?

- Exhibits no electrical resistivity under a critical temperature (T_c)
- Current known materials must be cooled to 9K (-260°C)

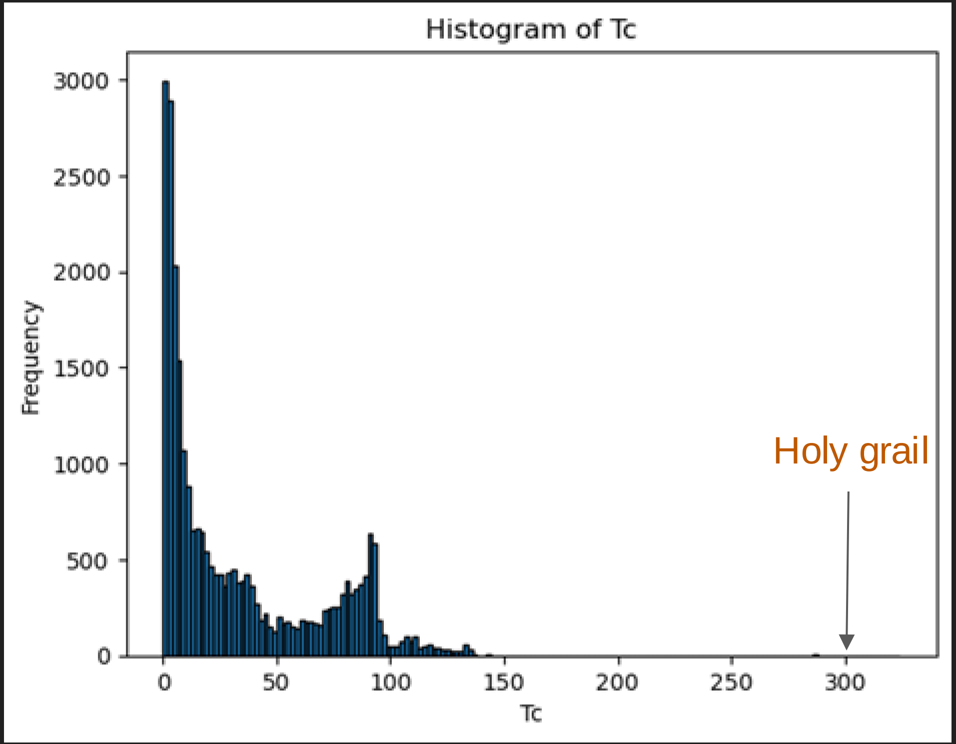
What are applications?

- High-speed rail
- MRI machines
- Microchips

Approach: Computational Materials Science

Databases: SuperCon

- Most extensive of its kind (26.000 entries)
- Records T_c for experimentally studied materials
- Many low-T_c, some high-T_c materials



MDR MATERIALS DATA REPOSITORY

Databases: SuperCon

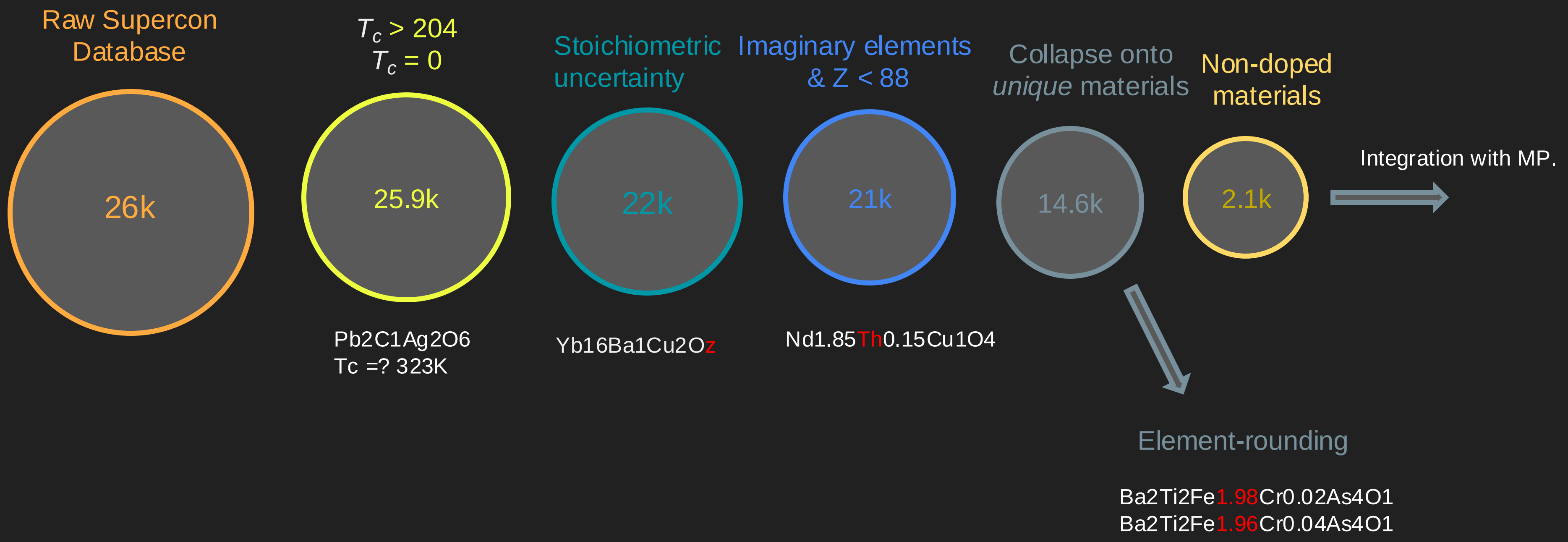
	data number	common formula	chemical formula	common name of structure	unit of Tc	Tc	journal
1	2	(Ba,La)2CuO4	Ba0.2La1.8Cu1O4-Y	T	K	29.0	Jpn.J.Appl.Phys., 26(1987)L223
2	3	(Ba,La,Ag)2CuO4	Ba0.1La1.9Ag0.1Cu0.9O4-Y	T	K	26.0	Jpn.J.Appl.Phys., 26(1987)L223
3	4	(Ba,La,)2CuO4	Ba0.1La1.9Cu1O4-Y	T	K	19.0	Jpn.J.Appl.Phys., 26(1987)L223

26.000 entries

Main drawbacks:

- Bad data!
  - Troves of missing values
  - T_c not recorded under same conditions
  - Human error in recording chemical formula
  - No descriptors

Databases: Pre-processing pipeline



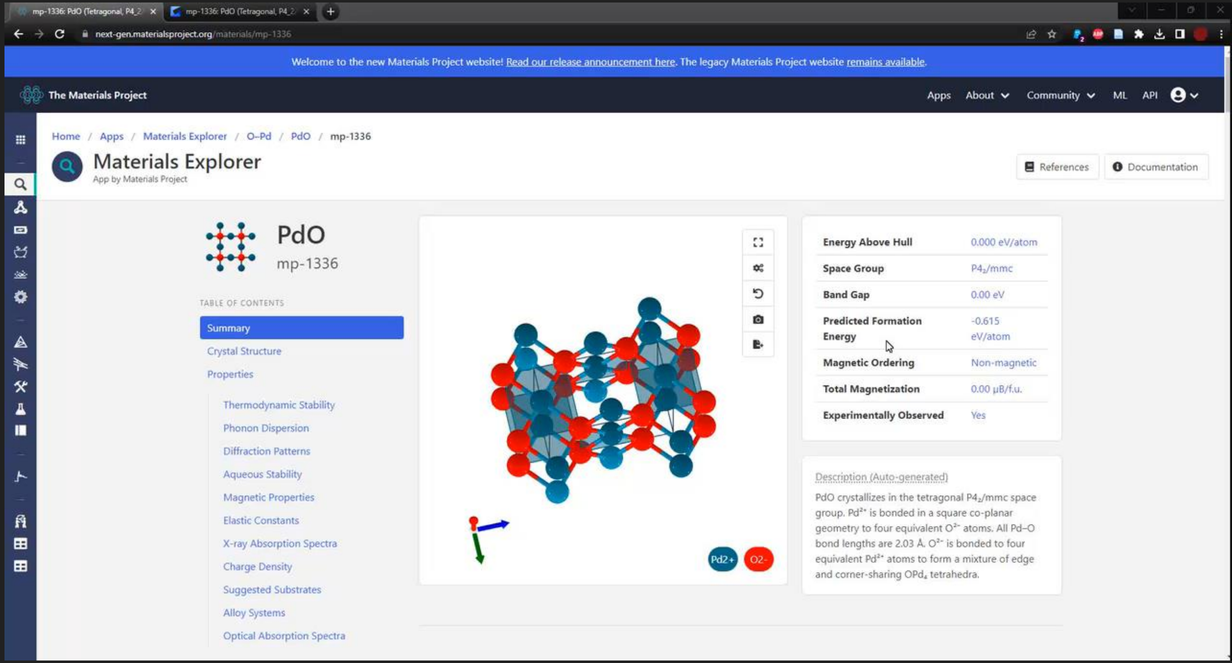
Databases: Matching materials with vast repositories

Density-Functional Theory Databases

https://next-gen.materialsproject.org/

Highlight: Materials Project:

- 150.000+ structures
- Lots of useful descriptors
- 800 matches with SuperCon



Machine learning: multi-stage pipeline

Does XYZ superconduct at all? *If so…* *At what temperature does XYZ superconduct?*

76.000 novel materials

Room-temp superconductor!

Model A:

- Classification
- Trains on 14k superconductors & 76k insulators
- Out-of-sample 94% accuracy

Model B:

- Regression on T_c
- Trains on T_c of 14k superconductors
- Out-of-sample *RMSE* of ±9.5K, $R^2 = 0.93$

Machine learning: 1) compositional descriptors

Features are statistics of the properties of constituent elements

Atomic Weight
First Ionization Energy
Covalent Radius
Density
Electron Affinity
Enthalpy of Fusion
Specific Heat Capacity
Number of Valence Electrons

Table: Elemental properties used in compositional models

Example: MgB_2

.	Magnesium	Boron
# element	1	2
# valence electrons	2	3

Feature: Weighted average of # Valence e- : **2.66**

What gives the model predictive power?

Tried different models:

- Gradient-boosted decision trees work best
- Decently interpretable, scalable, and fast

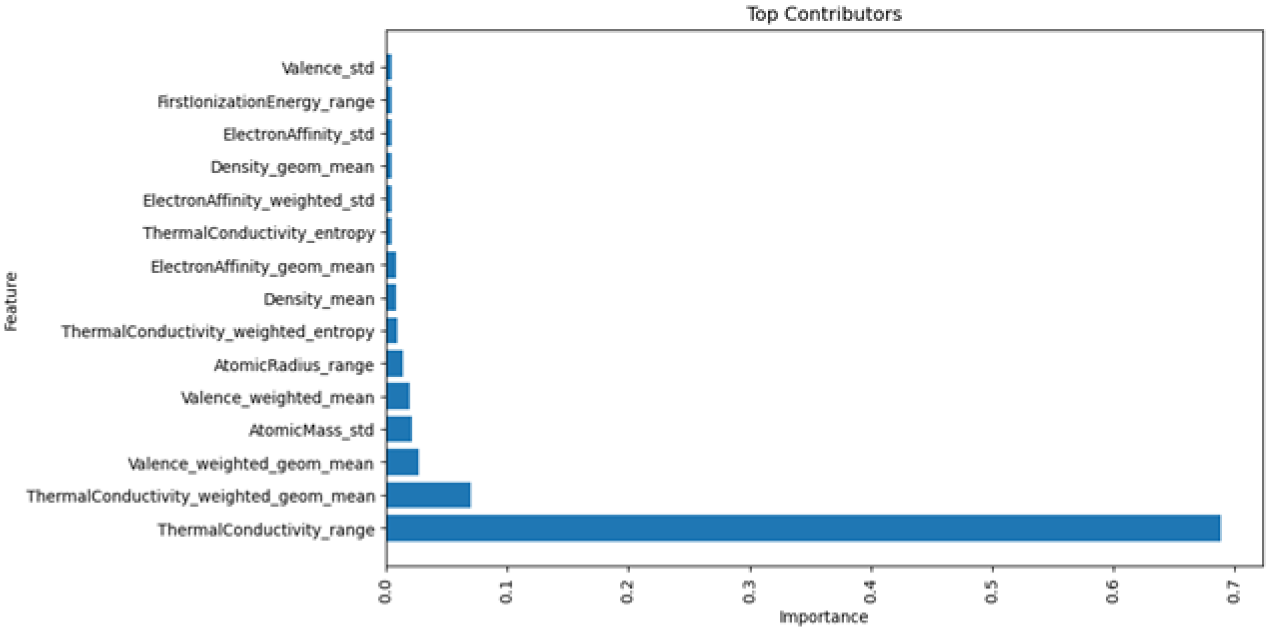
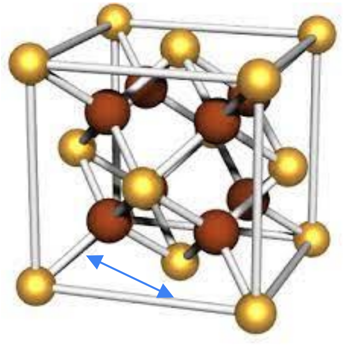
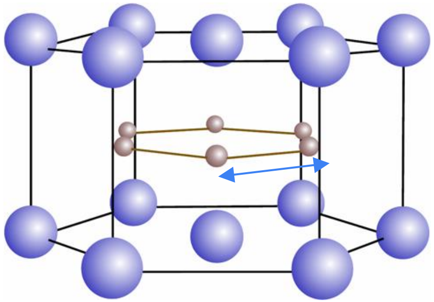


Fig: Feature-importances in regression model to predict Tc

Machine learning: 2) structural descriptors





Featurization from unit cell of the crystal:

- Vector of Minimum Relative Bond Distances
- State of the art: **Crystal-Graph Convolutional Neural Networks,** Xie and Grossman (2018)
- Limited by only having 800 matches so far

Machine learning: some previous results

Table 3. List of potential superconductors identified by the pipeline

Compound	ICSD	SYM
$CsBe(AsO_4)$	074027	Orthorhombic
$RbAsO_2$	413150	Orthorhombic
$KSbO_2$	411214	Monoclinic
$RbSbO_2$	411216	Monoclinic
$CsSbO_2$	059329	Monoclinic
$AgCrO_2$	004149/025624	Hexagonal
$K_{0.8}(Li_{0.2}Sn_{0.76})O_2$	262638	Hexagonal
$Cs(MoZn)(O_3F_3)$	018082	Cubic
$Na_3Cd_2(IrO_6)$	404507	Monoclinic
$Sr_3Cd(PtO_6)$	280518	Hexagonal
$Sr_3Zn(PtO_6)$	280519	Hexagonal
$(Ba_5Br_2)Ru_2O_9$	245668	Hexagonal
$Ba_4(AgO_2)(AuO_4)$	072329	Orthorhombic

Table: New materials confidently classified as superconducting (Stanev et. al, 2017)

Material	Predicted T_c (K)
$CsBe(AsO_4)$	13.7
$RbAsO_2$	8.0
$KSbO_2$	10.2
$RbSbO_2$	11.8
$CsSbO_2$	10.1
$AgCrO_2$	53.3
$K_{0.8}(Li_{0.2}Sn_{0.76})O_2$	18.6
$Cs(MoZn)(O_3F_3)$	20.5
$Na_3Cd_2(IrO_6)$	17.4
$Sr_3Cd(PtO_6)$	12.8
$Sr_3Zn(PtO_6)$	12.4
$(Ba_5Br_2)Ru_2O_9$	17.0
$Ba_4(AgO_2)(AuO_4)$	56.7

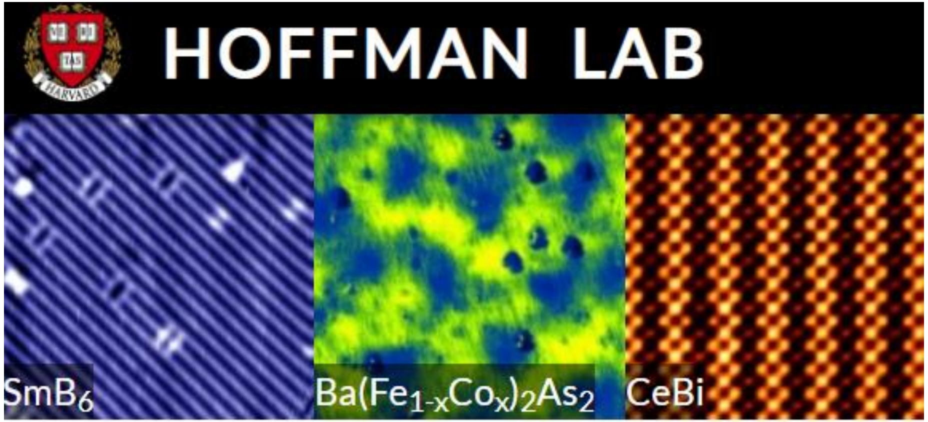
Table: Tc predictions for said list of potential superconductors (Hamidieh et. al, 2018)

Machine learning: Improvements

1. Approximate doped materials to parent compounds
2. What are more serviceable structural descriptors to use for ML?
3. Superconductivity depends sensitively on low-energy scale electronic state

Three take-aways

1) Finding a superconductor that functions at room-temperature would be *really awesome*
2) Approaches leveraging big data are promising and new
3) In exceptional materials discovery, computer-methods complement experiment nicely



Thank you!

Questions?